First Half of 2026 Freshmen Seminar (DFRC #51)

Freshmen Mock Case Scenario

Multi-platform Digital Forensics

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Keywords

Digital Evidence

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Freshmen Mock Case Scenario (DFRC #51)

Case Overview

- **Objectives**
  - Evaluation of basic techniques for digital data collection and analysis across various platforms
  - Inferring case scenarios and reconstructing timelines based on the provided digital evidence
- **Evaluation Factors / Scoring**
  - Evaluation Factors
    - Is the use of tools appropriate, and is the analysis process systematic and logical?
    - Has the scenario been logically reconstructed based on the digital evidence?
    - Was the report written clearly, and was the presentation easy to understand?
  - Scoring
    - Problem Solving (60 pts) + Drafting Timeline & Visual Summary (20 pts) + Presentation (20 pts)

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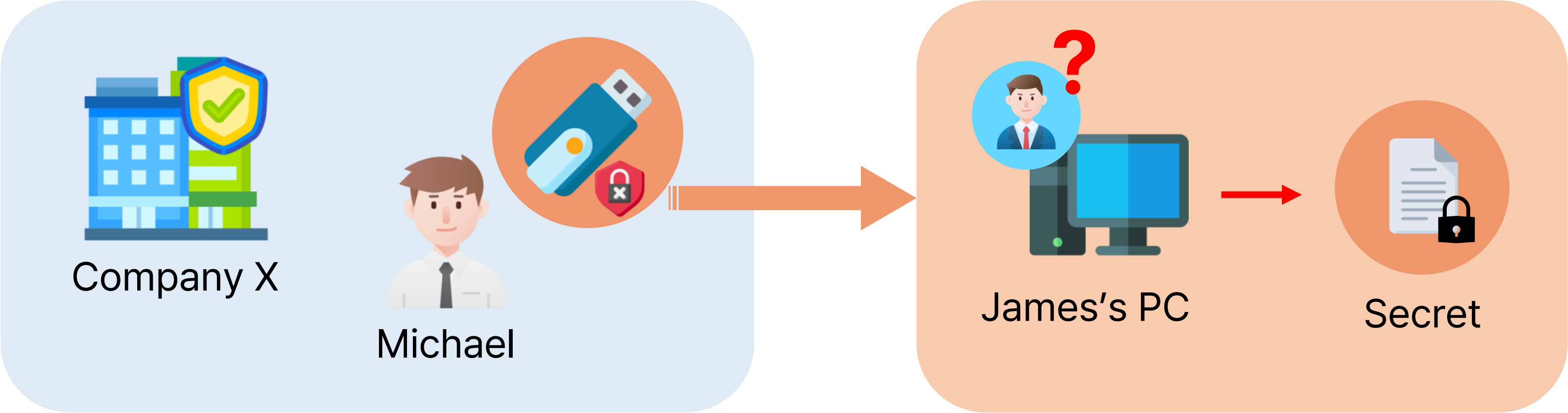
Case Overview

- **Mission: Reveal the truth behind the defense company X's technology leakage case!**
  - Michael, the employee of X, has been captured from carrying out unauthorized USB
  - During investigation, important files that only James can read were found in Michael's PC
  - Now, James is also suspected of being a suspect in the technology leakage case



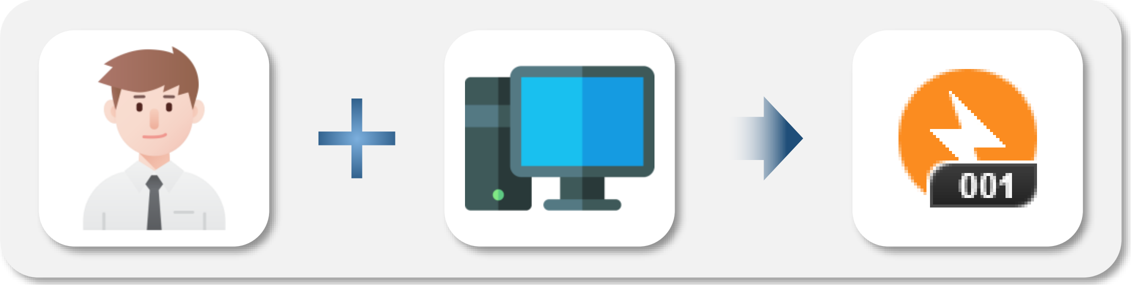
  - <u>You have to figure out the case's timeline, and reveal the truth behind of it!</u>

Freshmen Mock Case Scenario (DFRC #51)

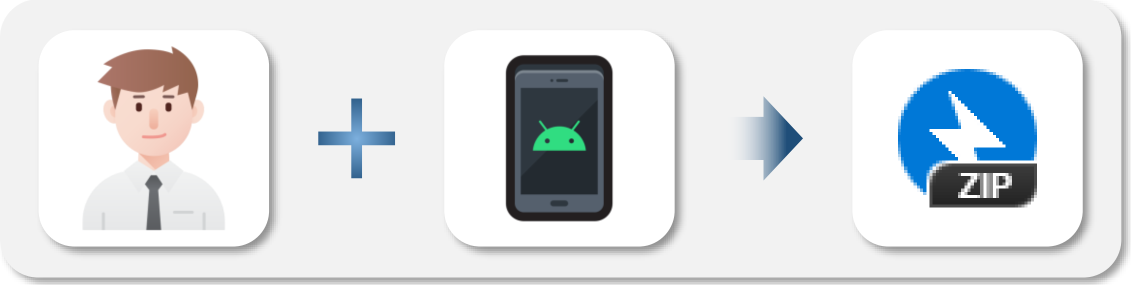
Case Overview

- **Investigation Materials**

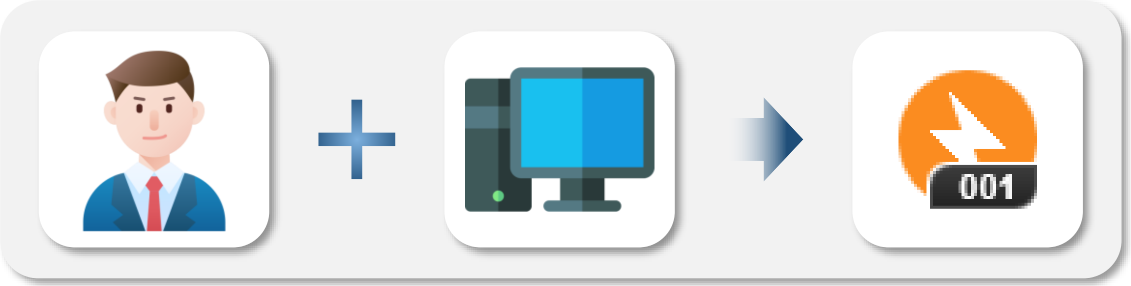
① Michael's PC (Michael_PC.001)



② Michael's Phone (Michael_Mobile.zip)



③ James' PC (James_PC.001)



④ AI Server PC (Server_PC.001)

Questions

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Questions

- **Q1. Based on the MMS conversation between James and Michael, identify and describe the following information obtained from the exchanged messages.**
  - Requirements
    - Daniel's business card information
    - James's PC password
  - Hint
    - The password for James' PC shared through MMS is stored in png format on Michael's Mobile.

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Questions

- **Q2. Describe the criminal plan and cost negotiation between Daniel and Michael based on the available digital evidence.**
  - Requirements
    - All communication methods used between Daniel and Michael
    - Name of the cryptocurrency exchange service used
    - Transaction ID
    - Michael's wallet address
    - Types of cryptocurrencies involved in the transaction
    - Relevant date and time of the transaction or exchange attempt
    - A detailed description of Daniel's criminal plan

Questions

- **Q3. Provide information about the AI server and where to get the information.**

Questions

- **Q4. Describe how Michael leaked data from the AI server.**

Questions

- **Q5. Create a table of who James contacted via Messenger and what conversations they had.**
  - Requirements
    - Describe everything you can find, including conversation content and attachments
  - Hint
    - Subkey : a2d9498030a9f8160265df7b9f87643e60d87c8130732a725b260200498cb614

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Questions

- **Q6. Summarize the case based on all evidence collected so far.**
  - Submit: ① Character details, ② Full incident timeline, ③ Visual summary of the case
    - Follow the templates as closely as possible and provide all necessary details (multiple pages allowed)

[① Character details] → Use below template

Name : Michael

Company :

Role :

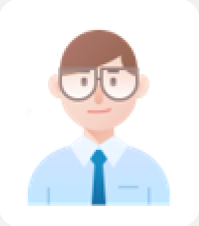
Artifacts :
- (ex) KakaoTalk (shared confidential files w/ friend)
- …

Name : James

Company :

Role :

Artifacts :
- (ex)



Name : Daniel

Company :

Role :

Artifacts :
- (ex)

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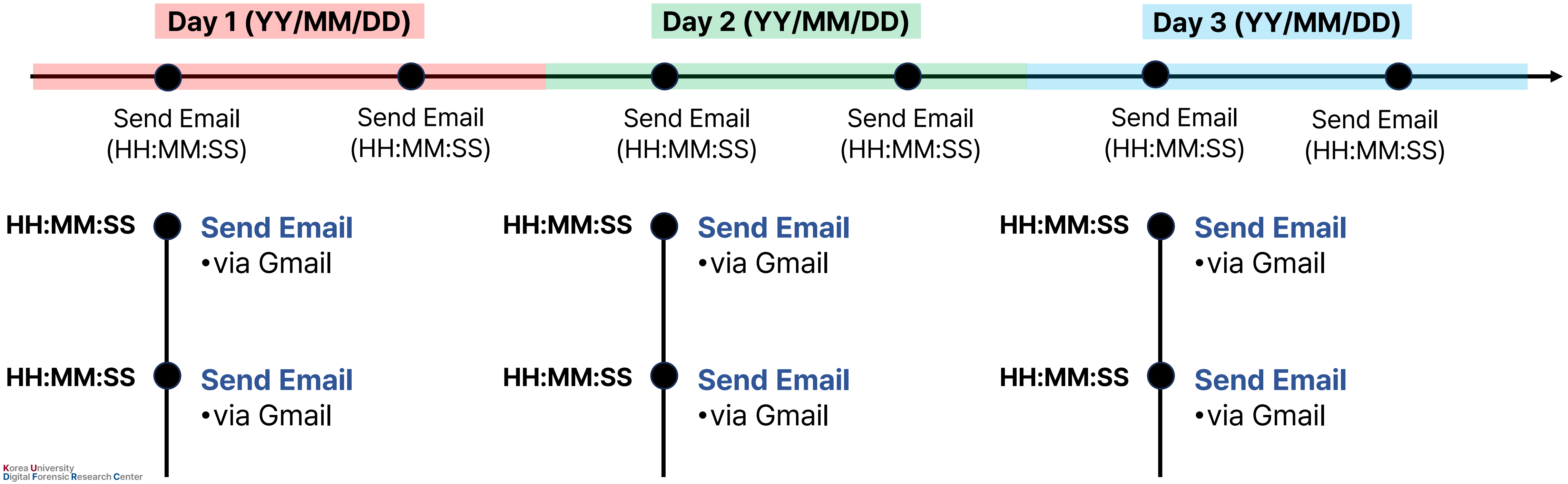
[② Full incident timeline] → Use below template



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[③ Visual summary of the case] → Diagram it freely ☺

Guidelines

Guidelines - Answer Preparation

- **Format**
  - Language: English (Solutions should be written in English only)
  - Timestamps: Use ISO 8601 format (KST) **ex.** 2026-01-08T21:40:00
- **Analysis & Reporting**
  - Clearly specify artifact paths in detail and include supporting evidence as much as possible
  - Do not use obtained credentials/session info for any online actions (ex. login, data modification)
  - Focus on main characters in the scenario (others can be disregarded)
- **Support**
  - Questions should be shared in KakaoTalk chat room
    - Feel free to ask any questions if you're struggling 🤓

Guidelines - Submission

- **Submission:** Solution PPT file
  - Follow DFRC lab PPT template (must install given font files)
  - Fill up the slides with your own solutions
  - Only the PDF file needs to be submitted (additional evidence files are not required)
- **Submission Deadline:** By 13:00 on January 27, 2026
  - Late submissions: 0 points
  - Prepare for answer presentations on January 27, 2026
- **Submmision Method:** (Email) ywpark0124@korea.ac.kr (석사 49기 박예원)
  - Mail subject form: [2026 상반기 DFRC 신입생세미나 모의케이스 답안 제출 / 본인 이름]

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Guidelines - Recommended Open Source Tools & References

- **Analysis Tools**
  - FTK Imager: Analyze image dump files & display folder/file tree structure
  - Autopsy: Open-source digital forensics suite for analyzing OS, file systems, and etc.
  - DB Browser for SQLite, SQLite Expert: SQL database file analysis
  - ALEAPP, iLEAPP: Smartphone image file analysis (Android/iOS)
- **Papers**
  - LangurTrace: Forensic analysis of local LLM applications
  - Decrypting messages: Extracting digital evidence from signal desktop for windows

THANK YOU
for Listening!

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Questions?